

Day-5

17. Find the average salary of instructors in the Computer Science department .

Query: select avg (salary)
from instructor
where dept_name= 'Comp. Sci.';
98133.470000

18. Find the total number of instructors who teach a course in the Spring 2018 semester.

Query:

```
SELECT COUNT(DISTINCT ID)
FROM teaches
WHERE semester = 'spring';
20
```

19. Find the number of tuples in the *course* relation.

Query:

```
SELECT COUNT(*) FROM course;
200
```

20. Find the average salary of each department.

Query:

```
SELECT dept_name,AVG(salary)
FROM instructor
GROUP by dept_name;
```

dept_name	AVG(salary)
Accounting	48716.592500
Astronomy	79070.080000
Athletics	77098.198000
Biology	61287.250000
Comp. Sci.	98133.470000
Cybernetics	96346.567500
Elec. Eng.	74162.740000
English	72089.050000
Finance	105311.380000
Geology	99382.590000
Languages	57421.856667
Marketing	84097.437500
Mech. Eng.	79813.020000
Physics	114576.900000
Pol. Sci.	100053.073333

Psychology	61143.050000
Statistics	67795.441667

21. Find the average salary of all instructors.

Query:

```
SELECT dept_name, AVG(salary)
FROM instructor;
77600.188200
```

22. Find the number of instructors in each department who teach a course in spring 2008 semester.

Query:

```
select dept_name, COUNT(DISTINCT id)
FROM instructor , teaches
WHERE instructor.id = teaches.id and semester = 'Spring' and year = '2008'
GROUP by dept_name;
```

AMBIGUOUS ERROR

23. Find the average salary of instructors in those department where the average salary is more than \$42000.

Query:

```
SELECT dept_name, AVG(salary) as avg_salary
from instructor
```

```
GROUP BY dept_name
```

```
HAVING AVG(salary) > 4200;
```

dept_name	avg_salary
Accounting	48716.592500
Astronomy	79070.080000
Athletics	77098.198000
Biology	61287.250000
Comp. Sci.	98133.470000
Cybernetics	96346.567500
Elec. Eng.	74162.740000
English	72089.050000
Finance	105311.380000
Geology	99382.590000
Languages	57421.856667
Marketing	84097.437500
Mech. Eng.	79813.020000
Physics	114576.900000
Pol. Sci.	100053.073333
Psychology	61143.050000
Statistics	67795.441667

24. Find total salary of the instructors.

Query:

```
SELECT SUM(salary)
FROM instructor;
3880009.41
```

25. Find the courses taught in the both the Fall 2007 and Spring 2008 semesters.**Query:**

```
SELECT DISTINCT course_id
FROM section
WHERE semester = 'Spring' and year = 2008
and
course_id in(
    SELECT course_id
    FROM section
    WHERE semester = 'Spring' and year = 2007);
course_id
362
852
991
962
237
349
345
158
704
```

26. Find all the courses taught in the both the Fall 2007 but not in the Spring 2008 semesters.**Query:**

```
SELECT DISTINCT course_id
FROM section
WHERE semester = 'Spring' and year = 2008
and
course_id not in(
    SELECT course_id
    FROM section
    WHERE semester = 'Spring' and year = 2007);
course_id
362
852
991
962
237
349
345
```

27. Name all instructors whose name is neither “Mozart” nor Einstein”.**Query:**

```
select distinct name
from instructor
where name not in ('Mozart', 'Einstein');
name
Lembr
Bawa
Yazdi
Wieland
DAgostino
Liley
Kean
Atanassov
Moreira
Gustafsson
Bourrier
Bondi
Soisalon-Soininen
Morris
Arias
Murata
Tung
Luo
Vicentino
Romero
Lent
Sarkar
Shuming
Konstantinides
Bancilhon
```

28. Find the total number of (distinct) students who have taken course sections taught by the instructor with ID 10101.**Query:**

```
select distinct name
from instructor
where name not in ('Mozart', 'Einstein');
name
Lembr
Bawa
Yazdi
```

Wieland
DAgostino
Liley
Kean
Atanassov
Moreira
Gustafsson
Bourrier
Bondi
Soisalon-Soininen
Morris
Arias
Murata
Tung
Luo
Vicentino
Romero
Lent
Sarkar
Shuming
Konstantinides
Bancilhon